Different Places, Different Colleges: A Geographic Examination of Local Postsecondary Landscapes

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Abstract

Geographic areas vary in their postsecondary institutional compositions. However, few studies have described the landscapes of higher education in geographic areas. We examine variations in college landscapes across geographic areas and their association with area characteristics. We find distinct typologies of colleges and geographic areas based on their college landscapes.

Introduction

According to the National Center for Education Statistics' most recent count of postsecondary institutions, there were 3,798 degree-granting colleges and universities in the United States in 2025-26 (NCES, n.d.). These institutions varied widely in their institutional characteristics, including the level of degrees offered, the types and number of academic programs available, the population of students served, and whether they were under public or private control. Institutional characteristics can also influence students' college choices (Skinner, 2019). And, although students frequently consider these characteristics simultaneously, researchers and policymakers often discuss different types of colleges based on these characteristics separately.

In addition to colleges' institutional characteristics, many students' college choices are also influenced by where colleges are located (Klasik et al., 2018; Toutkoushian et al., 2024). Most students attend college relatively close to their homes (Turley, 2009) and students are less likely to choose colleges that are farther from home (Jepsen & Montgomery, 2009; Long, 2004). Further research showed that the number of four-year institutions in a students' local area are positively associated with enrollment closer to home (Toutkoushian et al., 2024). Aligning with these findings, researchers have broadly characterized geographic areas based on the extent to which college opportunities are available within them (Dache-Gerbino, 2018; Hillman, 2016). While informative, these descriptors provide limited information about the types of colleges present within local areas.

Scholars have also examined how colleges' geographic locations are associated with the population characteristics of their local areas (Hillman, 2016; Klasik et al., 2018; Turley, 2009).

This literature highlights how the availability of different types of colleges varies for different demographic groups.

This study aims to extend prior research on spatial access to postsecondary opportunities by examining variations in the types of colleges that exist across geographic areas, and how areas with different configurations of postsecondary opportunities are associated with the areas' population characteristics. Using data from the Integrated Postsecondary Education Data System and the American Community Survey, we use a series of cluster analyses to identify types of postsecondary institutions and classify core-based statistical areas (CBSAs) based on the number and types of colleges located within them. We also use multinomial logistic regression to explore the extent to which population characteristics differ between different types of CBSAs. We do this to answer the following research questions:

1. How do the landscapes of postsecondary institutions differ between core-based statistical areas (CBSAs)?
2. To what extent are CBSA types associated with CBSA population characteristics?

Literature review

The Influence of College Proximity

Research on college access and choice has generally shown a significant relationship between college enrollment and the proximity of postsecondary institutions (Frenette, 2006; Jepsen & Montgomery, 2009; Toutkoushian et al., 2024). The positive association between institutional proximity and college enrollment is explained by the concept of "distance elasticity" (Hillman & Weichman, 2016). According to Turley (2009), college proximity may influence students' college-going through "predisposition" and "convenience" mechanisms. Particularly, students are more likely to enroll in college when there are college campuses near their homes

because they may feel more comfortable with higher education as a result of having institutions in their local areas (predisposition). The presence of proximal colleges may also ease students' transitions to college logistically, financially, and emotionally. In short, students' local environments may shape their college opportunities and choices based on the presence (or lack thereof) of postsecondary institutions.

Defining College Geographies

The number and types of colleges vary between geographic areas. Researchers have described geographic areas using terms such as *deserts* and *oases* based on the presence and relative number of postsecondary institutions in areas of interest (Dache-Gerbino, 2018; Hillman, 2016). The term *education deserts* provides a bit of nuance to geographic descriptors by accounting for institution level. Education deserts are defined as areas with no postsecondary institutions or where the only broad access institution is a two-year college (Hillman & Weichman, 2016). However, institutions differ from each other on a number of characteristics.

As stated, in studies of the geography of college opportunities, researchers typically classify institutions by whether they offer two- or four-year degrees and whether they are public or private (Hillman, 2016; Turley, 2009). Yet, colleges vary greatly in their institutional characteristics. The Carnegie Classification of Institutions of Higher Education categorizes colleges according to various characteristics, including their size, types of degrees awarded, and academic program focus (American Council on Education., *n.d.*). Indeed, Dache-Gerbino (2018) included descriptors of institutions' Carnegie classifications in her study of local college opportunities, while Hillman et al. (2021) examined different types of colleges within rural areas.

The distribution of college opportunities is influenced by historical and geographical reasons. For instance, Historically Black Colleges and Universities are predominantly located in

the Southern United States, reflecting the historical concentration of Black populations and the legacy of racial segregation in higher education (Allen & Jewell, 2002). Additionally, reflecting their missions to serve their local communities, a large proportion of postsecondary institutions in rural areas are two-year colleges (Hillman et al., 2021). This suggests that the local landscapes of geographic areas may be influenced in ways that cannot be accounted for by only examining colleges' degree award levels and institutional control.

Nuancing Local College Landscapes

Recent studies represent an important step toward understanding the geography of higher education. Ehlenz and Mawhorter (2022) conceptualized a typology of metropolitan areas based on the location and distribution of colleges within them. They classified the U.S. metropolitan areas into higher institution metros according to the number of higher education institutions and the dominance of the largest institution in each area. However, this analysis is limited to metropolitan statistical areas (excluding micropolitan statistical areas). Furthermore, the study does not account for the structure that institutions are nested in geographic areas.

Studies primarily focused on geographic proximity or broad institutional types offer a limited view of how opportunity is structured across regions. Therefore, less is known about how multiple institutional and regional characteristics interact to shape the structure of higher education opportunity across broader geographic contexts. By integrating both institutional- and regional-level characteristics, the analysis that systematically captures dimensions of institutional distribution and stratification, such as minority-serving institution- status, level, and institutional type, is necessary. The focus is on how the structure of higher education opportunity varies across regions, emphasizing structural disparities in institutional distribution rather than classifying regional types.

Conceptual Framework

Colleges are complex institutions that vary along many dimensions of institutional characteristics. Similarly, geographic areas contain varying types and numbers of postsecondary institutions. Considering local postsecondary landscapes are constructed by institutional and geographical characteristics, any typology of them should encompass both aspects. As such, we use two concepts to examine CBSAs' postsecondary landscapes.

First, we describe institutions using a conceptual framework which Harris (2020) used to create a typology of U.S. colleges and universities. In this framework, Harris identified four key aspects of institutional diversity: systemic, programmatic, procedural, and constituential. Systemic diversity accounts for institutional structure and governance. It describes whether an institution is a two- or four-year college or university, whether it is public or private, and the number of students enrolled. Programmatic diversity describes colleges' academic offerings, including the level of awards it confers, as well as the number and variety of degree programs offered, which are fundamentally related to institutions' academic programs or educational mission. Procedural diversity refers to the ways colleges offer and implement programs, and reflects qualities such as whether colleges offer online, evening, or weekend education programs, how much money they allocate to instructional costs, and how much they charge for tuition. Lastly, constituential diversity accounts for differences in the types of students colleges serve. Focusing on student characteristics, constituential diversity describes colleges' student populations based on their demographic characteristics, academic preparation, and enrollment level. Together, these four aspects of institutional diversity provide a clear picture for understanding how colleges and universities differ from one another which can then be used to examine how the number and types of colleges vary between geographic areas.

We use the concept of geography of opportunity to guide our examination of differences in CBSA postsecondary landscapes. This concept focuses on spatial variations in the presence, perceptions of, and access to opportunities within geographic areas (Briggs, 2003; Galster & Killen, 1995; Reece & Gambhir, 2009; Tate, 2008). In the field of higher education research, geography of postsecondary opportunities highlights how geographic areas differ based on the number and types of colleges and universities located within them, and has been used to examine college access by scholars such as Turley (2009), Hillman and Boland (2018), Dache-Gerbino (2018), and Reyes et al. (2019). Following the work of Hillman and Boland (2018) and Dache-Gerbino (2018), geography of opportunity helps us to create a typology of CBSAs to describe the differences in postsecondary opportunities between using the institutional typology we develop in this paper.

Methods

Data

We obtained data for this study from three sources. We gathered data on postsecondary institutions reflecting their systemic, programmatic, procedural, and constituential characteristics from the 2024 Integrated Postsecondary Education Data System (IPEDS) survey. Hispanic-serving institutions were identified using the Carnegie Classification of Institutions of Higher Education 2025 Public Data File.[1] Data on each CBSA characteristics, including total population, median income, racial/ethnic composition, percentages of individuals with a bachelor's degree or higher, and percentages living below poverty level were collected from the U.S. Census Bureau's 2024 American Community Survey five-year estimates.

Sample

[1] We used 2025 public data since the two most recent Carnegie Classification public data were 2021 and 2025.

The sample for this study consisted of degree-granting and Title IV-eligible, public and not-for-profit private two- and four-year postsecondary institutions in the 50 U.S. states and Washington, DC. Only colleges and universities that were located within a metropolitan or micropolitan statistical area were included in the sample due to the study's focus on describing the characteristics of CBSAs. Additionally, because the study focuses on the enrollment destinations of degree-seeking undergraduate students, we excluded institutions that primarily awarded certificates or graduate-level degrees, as well as those that only offer distance education programs. While we retained most special focus institutions, such as those that focused on arts and sciences, medical and health sciences, business, and career studies, we excluded those theological colleges due to their narrow program focus. Our final sample consisted of 2,380 institutions in 719 CBSAs.

Variables

The variables we included in our analyses were chosen to align with Harris' (2020) institutional diversity framework. To address colleges' systemic diversity, we accounted for institution type (e.g., whether it was a public or private two-year college or a public or private four-year college) and an indicator of whether the institution was a part of a college system. Variables reflecting colleges' programmatic diversity included their Carnegie award level focus and special focus classifications as well as the number of degree programs they offered. To account for colleges' procedural diversity, we included their proportion of online programs, an indicator of whether they offered programs that could be completed in the evening or on weekends, their instructional expenditures per full-time equivalent student, and their in-state tuition cost. Constituential diversity was represented by each college's admissions rate (measured by the proportion of applicants that were admitted), total student population, status as a

minority-serving institution, their proportions of students who were adult, Pell grant recipients, non-White, and enrolled in degree-granting undergraduate programs.

Variables included in the regression analysis reflected the population characteristics of colleges' CBSAs. These variables were the total population, population density, percentage of adults with at least a bachelor's degree, median income, the percentages of residents living below the poverty level and who were unemployed, and the percentage of residents by race/ethnicity (White, Black, Hispanic, or Asian).

Analytic Approach

In this study, we adopt a two-stage analytical approach to establish a typology of core-based statistical areas (CBSAs) based on the characteristics of postsecondary institutions located within them. In the first stage, we classify individual institutions into groups based on their institutional characteristics. In the second stage, we classify CBSAs based on the types and numbers of institutions located within each area. This approach allows us to examine differences in local postsecondary landscapes by considering both individual institutional characteristics and the composition of institutions within CBSAs.

In the first stage, we identified groups of postsecondary institutions based on their shared characteristics. To do this, we conduct a principal component factor analysis, specifically a factor analysis of mixed data (FAMD), to reduce dimensionality among the 16 variables that reflected institutional diversity according to Harris' (2020) typology and to accommodate the continuous and categorical variables. FAMD also allows us to summarize shared patterns and reduce redundancy among correlated institutional characteristics (Pagès, 2004). We use the dimensions to construct a dissimilarity matrix which includes calculated Euclidean distances between institutions. We then use this dissimilarity matrix to perform hierarchical agglomerative

clustering (HAC) using Ward's linkage method. To evaluate alternative clustering solutions, we compare average silhouette widths and cluster sizes across solutions ranging from two to twelve clusters. The nine-cluster solution yields the highest average silhouette width among the institutional clustering solutions examined. Accordingly, we retain nine institutional clusters for the subsequent analysis.

In the second stage, we use HAC to classify CBSAs based on the composition of postsecondary institutions located within them. Using the nine institutional clusters identified in the first stage, we calculate the proportion of institutions belonging to each cluster within each CBSA. These nine institutional cluster counts serve as indicators of the local postsecondary landscape. We calculate Bray–Curtis dissimilarities between CBSAs based on these counts and perform HAC using Ward's linkage method. We examine average silhouette widths and cluster sizes across alternative solutions and retain an eight-cluster solution for subsequent analyses. This approach allows us to identify groups of CBSAs with similar institutional compositions and examine differences in local postsecondary landscapes across the United States.

Finally, we conduct multinomial logistic regression to examine the relationships between CBSA characteristics and CBSA cluster membership. The dependent variable is membership in one of the eight CBSA clusters, with Cluster 1 serving as the reference category. The independent variables include total population, population density, educational attainment, median household income, poverty rate, unemployment rate, and the percentages of non-Hispanic Black, Hispanic or Latino, and non-Hispanic Asian residents. All continuous predictors are standardized before estimation to facilitate comparisons across variables. We report estimated regression coefficients, standard errors, and statistical significance to assess the associations between CBSA characteristics and cluster membership.

Limitations

There are two key limitations to this study. While research suggests that students are more likely to attend for-profit colleges compared to community colleges when they are located within their local areas (Lee & Pirog, 2023), we do not account for for-profit colleges in this study. This is because including these colleges limits our ability to examine the geographic distribution of other types of institutions due to sheer number and prevalence of for-profit colleges. Second, findings of agglomerative hierarchical clustering must be interpreted with care. HCA is predictive, not deterministic. This means that group membership within any given cluster is predicted through an algorithm based on the provided data and classification errors are possible. Despite this, the results of the analyses show clear patterns that can be used to identify CBSA types.

Results

Table 1 contains descriptive statistics for the institutions and CBSAs, and depicts the wide variation in characteristics among analytic units at both levels. Among postsecondary institutions, we see that most colleges were four-year or public two-year colleges and only 2% were private two-year colleges. However, IPEDS classified community colleges that offer bachelors degrees as four-year institutions, the distribution of institution types did not fully align with the distribution of colleges' award level focus Carnegie classification, which showed that 41% of colleges primarily awarded associates degrees. Most colleges were classified as not special focus or professions-focused, meaning that most of the degrees they offered were pre-professional or career-aligned (ACE, n.d.). Another notable characteristic was the relatively high mean admissions rate among the colleges. Among the CBSAs, there were similar

proportions of micropolitan and metropolitan statistical areas (48% vs. 52%, respectively) with an average of three colleges among all CBSAs.

Agglomerative Cluster Analyses

We display the eigenvalues, variance explained, and variables that contributed most for the first 12 dimensions resulting from the FAMD. We retained the first 10 dimensions because their eigenvalues were greater than one and their cumulative variance explained was 69%, in accordance with the Kaiser criterion (Kaiser, 1960). Using the first 10 dimensions, we calculated a dissimilarity matrix and used the results to perform HCA. Table 3 shows the average silhouette widths and the numbers of observations in the smallest (Min_n) and largest (Max_n) clusters for each clustering solution at both the institution and CBSA levels. Based on the results shown in the first panel of the table, we determined that there were nine groups of institutions because they had the highest mean silhouette score. Although the score of .295 indicates weak separation between groups, visual inspection of the data showed logical groupings. Furthermore, this cluster solution reflected the complexity of postsecondary institutions and provided a reasonable basis for hierarchical clustering of CBSAs, which was the next step of our analysis. We provide descriptions and examples of colleges within each group in Table 4.

The second panel of Table 3 shows the cluster validation for the second HCA. The silhouette scores increased across the alternative CBSA clustering solutions examined, with the highest score observed for the twelve-cluster solution. However, based on our knowledge of higher education institution characteristics and the interpretability of the clusters, we selected the eight-cluster solution (Teh et al., 2018). This solution had an average silhouette width of .569.

Table 5 summarizes the distribution of institution types across the CBSA clusters as well as the types of postsecondary institutions located within them in Panel A. Descriptive titles for each of the CBSA clusters and names of example CBSAs are listed in Panel B.

Cluster 1: Singular Baccalaureate-Dominant Institution Areas. This cluster consisted of 88 CBSAs, each of which only contained a single public two-year college or low-cost four-year college. Some CBSAs included in this group were Amherst Town-Northampton, MA and Athens, TN.

Cluster 2: Minimal Opportunity, Broad-Access Anchor Institution Areas. There were a total of 191 CBSAs and 248 colleges in this cluster. Over 90% of the institutions were public two-year and and broad-access four-year colleges. The Hagerstown-Martinsburg, MD-WV and Wilmington, NC CBSAs were included in this cluster.

Cluster 3: Extensive-Opportunity Higher Education Ecosystems. The 101 CBSAs in this cluster contained a total of 341colleges; 47% of which were bachelor's degree-focused institutions that awarded most degrees in professions-focused as well as arts and sciences programs. Two- and four-year broad access institutions were also prevalent in this cluster, making up 21% of the institutions. This cluster included the Cincinnati, OH-KY-IN and Greenville-Anderson-Greer, SC CBSAs.

Cluster 4: Singular University Hubs. This cluster had 71 CBSAs where the only postsecondary option was generally only one large, graduate degree-granting university. Sample CBSAs for this cluster were Bozeman, MT and Troy, AL.

Cluster 5: Broad Access and MSI-Opportunity Areas. Eighty-nine CBSAs were classified as a part of this cluster. These CBSAs contained a total of 209 institutions, with 40% being public two-year or low-cost four-year colleges, 33% being large, graduate universities, and

17% of the institutions being four-year historically Black colleges and universities and Hispanic-serving institutions. There were also seven tribal colleges and universities in this cluster, the highest of all clusters. Albuquerque, NM and Knoxville, TN were two of the CBSAs included in this cluster.

Cluster 6: Robust Higher Education Markets. The 128 CBSAs in this cluster contained the greatest number of colleges: 1,301 institutions. Colleges of all types were found in this cluster, most of which were public, two- and four-year broad-access institutions (27%) and private four-year institutions (26%). Unsurprisingly, CBSAs including the Los Angeles-Long Beach-Anaheim, CA, New York-Newark-Jersey City, NY-NJ, and Washington-Arlington-Alexandria, DC-VA-MD-WV areas were in this cluster.

Cluster 7: Private College Hubs. This was a relatively small cluster with only 32 institutions nested within 23 CBSAs. Nearly all (91%) of the colleges were private, four-year institutions. Examples of CBSAs found in this cluster were the Bridgeport-Stamford-Danbury, CT and Sioux Falls, SD-MN areas.

Cluster 8: Two-Year and Career-Focused Opportunity Areas. Two-year colleges that focused on applied and career studies (43%) were the most common colleges found in Cluster 8, which contained 28 CBSAs and 68 postsecondary institutions. The Beaumont-Port Arthur, TX and Bellingham, WA CBSAs were classified within this cluster.

The map in Figure 1 shows the locations of all of the CBSAs in our sample. Each CBSA is shaded based on its predicted cluster. The map shows how CBSAs' postsecondary landscapes differ across the country. For example, many of the CBSAs in the east, spanning from southern Maine to Virginia, are classified as Robust Higher Education Markets, reflecting that they contain a large number and wide variety of postsecondary institutions. As another example, the

map suggests that CBSAs in Arizona primarily offer broad access institutions since most of them are shaded in purple and light blue. Hawai'i also offers an informative picture. There, the islands of Hawai'i and Oahu are classified as Broad Access and MSI-Opportunity Areas and Robust Higher Education Markets, respectively, while the other CBSAs are labeled as Minimal Opportunity, Broad Access Anchor Institution Areas.

Multinomial Logistic Regression Analysis

Table 6 displays the descriptive statistics for each CBSA cluster. The data show that the distribution of CBSAs by CBSA type differs across groups. For instance, nearly all CBSAs in Cluster 6 (Robust Higher Education Markets) are metropolitan statistical areas, meaning that they contain at least one urban area of 50,000 or more residents. The majority of CBSAs in Clusters 3, 5, and 8 are also metropolitan statistical areas. On the other hand, most CBSAs in Clusters 1, 2, 4, and 7 were micropolitan statistical areas, which have at least one urban area with a population less than 50,000 residents but greater than 10,000 residents. There was substantial variation in the mean number of colleges across CBSA types. CBSAs in Cluster 6 contained the highest number of colleges, 10.16 institutions on average. This was much more than the mean of the CBSA cluster with the next highest mean number of colleges Cluster 3 (Extensive-Opportunity Higher Education Ecosystems) with an average of 3.38 colleges. The number of institutions corresponded to the mean population and population densities of their respective CBSA clusters.

We conducted a multinomial logistic regression analysis to examine the extent to which CBSA clusters had significantly different population characteristics. Table 7 presents the results of the regression. In the analysis, Cluster 1 (Singular Baccalaureate-Dominant Institution Areas) was designated as the reference group. After controlling for other variables, total population size

showed a highly statistically significant positive relationship with the relative log-odds of belonging to Clusters 2, 3, 5, 6, and 8. Indeed, our results indicate that population size is one of the most consistently significant predictors of CBSA cluster membership. For example, the largest positive regression coefficient found in Cluster 6 (Robust Higher Education Markets) meant that a one-standard-deviation increase in total population was associated with a 12.06 increase in the log-odds of a CBSA being classified in Cluster 6 compared to Cluster 1. In contrast, population density was a significantly negative predictor of classification in Cluster 2 (Minimal Opportunity, Broad-Access Anchor Institution Areas, $\beta = -1.67$) compared to Cluster 1. This indicates that higher population density was associated with lower relative log-odds of belonging to Cluster 2 rather than Cluster 1, holding other variables constant. Conversely, population density was a significantly positive predictor of membership in Cluster 7 (Private College Hubs, $\beta = 1.23$) compared to Cluster 1, indicating higher relative log-odds of belonging to Cluster 7 rather than Cluster 1 as population density increased.

Differences were also observed across CBSA clusters in terms of educational attainment and household income. The proportion of residents with a bachelor's degree or higher showed a significant positive relationship with the relative log-odds of belonging to Clusters 4, 5, 6, and 8, with the largest positive regression coefficient observed in Cluster 4 (Singular University Hubs) ($\beta = 1.77$, $p < .001$). In contrast, a significant negative relationship was observed in Cluster 2 ($\beta = -0.66$, $p < .05$). Median household income showed a significant negative relationship in Clusters 3, 4, 5, 6, and 8, with the largest negative regression coefficient found in Cluster 4 ($\beta = -1.91$, $p < .001$). This suggests that, after controlling for other regional characteristics, educational attainment and household income may be associated with CBSA cluster membership in opposite directions.

Racial and ethnic composition also showed significant relationships with certain CBSA clusters. The proportion of non-Hispanic Black residents showed a significant positive relationship with the relative log-odds of belonging to Clusters 5 (Broad Access and MSI-Opportunity Areas), 6 (Robust Higher Education Markets), and 8 (Two-Year and Career-Focused Opportunity Areas). The proportion of Hispanic or Latino residents showed a significant positive relationship in Clusters 2, 4, and 5, with the largest regression coefficient observed in Cluster 5 ($\beta = 1.22$, $p < .001$). In contrast, the proportion of Asian residents showed no statistically significant relationship with membership in any of the clusters. In addition, neither the poverty rate nor the unemployment rate showed a significant relationship with cluster membership.

In summary, CBSA cluster membership based on the composition of higher education institutions was significantly associated with regional population size, educational attainment, median household income, and certain aspects of racial and ethnic composition. These results suggest that differences in higher education environments across regions are related not only to differences in the number of postsecondary institutions but also to regional population and socioeconomic characteristics.

Discussion

The vast number and wide variety of colleges and universities in the United States are unevenly distributed across the country (Hillman & Weichman, 2016). While higher education scholars have highlighted this spatial variation in college access across geographic regions (Dache-Gerbino, 2018; Ehlenz & Mawhorter, 2022; Hillman & Weichman, 2016), the resulting characterizations have not fully captured colleges' multidimensional attributes. Our study aimed to create a typology of CBSAs to reflect the number and types of postsecondary institutions

located within them. Our analyses resulted in the identification of eight types of CBSAs, including areas that contain only one four-year college and areas where private colleges are the predominant type of institution.

Our work contributes to the existing literature on the spatial distribution of higher education institutions by accounting for the many different institutional characteristics that can be used to describe colleges. Using Harris' (2020) framework for describing institutional diversity, we accounted for 16 different institutional characteristics in our analyses. In doing this, we were able to acknowledge important differences between institutions beyond the degrees they award and the size of their student populations to also capture qualities, such as the programs they offer and the types of students they serve. Furthermore, it is important to consider a wide range of characteristics because they play an important role in shaping students' college choices (Perna, 2006).

Accounting for many institutional characteristics allowed us to identify CBSA groups that reflected variation in the types of postsecondary opportunities available in each area. For example, we identified Robust Higher Education Markets that contained a large number and wide variety of institutional types as well as Singular University Hubs where the sole institution of higher education is a large university. These classifications merge, nuance, and provide additional information to descriptions of geographic areas posed by other scholars. This can be seen in our identification of Robust Higher Education Markets. According to Dache-Gerbino's (2018) work, these CBSAs would be considered college oases based on the presence of a large number of colleges that vary in their sector and award levels. These CBSAs would also be labeled as oases according to Hillman and Boland's (2018) typology based on the number of colleges and their accessibility in terms of admissions. In addition to acknowledging the number

of institutions in Robust Higher Education Markets, our typology reflects the presence of all types of colleges, including STEM-focused universities, minority-serving institutions, and career-focused two-year colleges. Another example of the alignment between our typology and prior work is the cluster we called Minimal Opportunity, Broad-Access Anchor Institution Areas. CBSAs of this type would likely be classified as refuges based on Hillman and Boland's (2018) typology due to the high level of accessibility of the community colleges and broad access four-year institutions, as well as the small number of colleges present in the areas.

As stated by Turley (2009), "neither the location of colleges nor the people who live near them are random" (p. 142). Therefore, in addition to classifying CBSAs, we also examined the relationship between CBSA cluster membership and CBSA population and socioeconomic characteristics to address the relationship between various college landscapes and local population characteristics (Dache-Gerbino, 2018). In accordance with prior findings, our results showed a significant association between the different types of CBSAs and the areas' population characteristics. Unsurprisingly, CBSAs' population sizes were highly associated with their classification. This is likely because areas with large populations are able to support the presence of many postsecondary institutions through a consistent supply of students. We also found that population density was associated with two types of CBSAs: Minimal Opportunity, Broad-Access Anchor Institution Areas (Cluster 2) and Private College Hubs (Cluster 7). Population density is negatively associated with the likelihood of CBSAs in being classified in Cluster 2. This cluster has a small number of colleges (particularly community colleges) that may serve a large area, suggesting that the population they serve are relatively dispersed. On the other hand, CBSAs in Cluster 7 are more likely to have higher population densities. This may be associated with the presence of broad access universities. Universities are typically larger than

community colleges meaning that they may have more faculty and staff which would be associated with a higher population density. Overall, these findings align with the concept of geography of opportunity by showing how the availability of higher education varies across CBSAs.

Conclusion

Research on the geography of college opportunities have acknowledged how the locations of postsecondary institutions are not evenly distributed across geographic areas and are associated with the population characteristics of the areas in which they are located. This study extends existing literature in the field by providing a more nuanced understanding of how college opportunities are distributed across the United States.

Findings from this study provide a new way of understanding of the geographic distribution of postsecondary institutions that can be informative for future research. In terms of future research directions, this typology of postsecondary landscapes can provide a lens through which researchers can examine student college application and enrollment behaviors. More specifically, including this data in future analyses of college-going may help to increase our understanding of college enrollment rates and decisions, particularly among underrepresented student populations. This is important because the colleges within students' local areas are associated with their likelihood of applying to and enrolling in college (Klasik et al., 2018). In short, this typology can guide further research to understand better how college-going is related the geography of students' local areas.

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Tables and Figures

Table 1. Descriptive statistics

Variable	Mean (SD)
Level 1: Postsecondary institutions (*N* = 2,380)	
Institution Type	
Public 2-year	0.27
Private 2-year	0.02
Public 4-year	0.30
Private 4-year	0.40
System Membership	
Not in system	0.60
In system	0.40
Carnegie Award Level Focus	
Associate	0.41
Baccalaureate	0.23
Master's	0.15
Doctorate	0.21
Special Focus	
Not special focus	0.39
Professions-focused	0.42
Applied and career studies	0.03
Arts and sciences	0.10
Business	0.01
Medical/health sciences	0.04
STEM	0.01
Number of Academic Programs Offered	96.57 (84.23)
Proportion of Programs Offered Fully Online	0.13 (0.19)
Weekend/Evening Course Availability	
No	0.42
Yes	0.58
Instructional Expenditures per FTE Student ($)	11,211.00 (9,653.33)
In-District Undergraduate Tuition ($)	18,285.42 (18,638.58)
MSI Status	
None	0.76
HBCU	0.04
HSI	0.19
TCU	0.01
Proportion of Adult Students	0.21 (0.17)
Proportion of Pell Grant Recipients	0.33 (0.14)
Proportion of Non-White Students	0.50 (0.23)
Total Student Enrollment	9,252.64 (13,598.30)
Proportion of Undergraduate Students	0.88 (0.17)

Variable	Mean (SD)
Admission Rate	0.84 (0.22)
Level 2: Core-based statistical areas (*N* = 719)	
CBSA Type	
Micropolitan statistical area	0.48
Metropolitan statistical area	0.52
Number of Postsecondary Institutions	3.31 (6.67)
Total Population (100,000s)	4.26(12.65)
Population Density (Persons per Square Mile)	190.25 (278.94)
Population Aged 25+ with a Bachelor's Degree or Higher (%)	28.60 (9.30)
Median Household Income ($)	69,645.57 (14,605.59)
Families Below the Poverty Level (%)	9.81 (3.78)
Unemployment Rate (%)	2.92 (0.84)
Non-Hispanic White Population (%)	69.88 (18.62)
Non-Hispanic Black Population (%)	9.01 (11.88)
Hispanic or Latino Population (%)	13.04 (15.09)
Non-Hispanic Asian Population (%)	2.30 (3.54)

Table 2. Dimensions Extracted from FAMD

Dimension	Eigenvalue	Explained Variance (%)	Cumulative Variance (%)	Top Contributing Variables
1	4.52	16.76	16.76	Tuition, Institution type, Award level
2	2.93	10.86	27.62	Number of degree programs, Size, Award level
3	2.08	7.7	35.32	Pell, Proportion of non-White students, MSI
4	1.9	7.04	42.36	Special focus, MSI, Proportion of non-White students
5	1.57	5.8	48.16	Special focus, Institution type, Award level
6	1.33	4.94	53.1	Institution type, Special focus, MSI
7	1.13	4.17	57.28	Special focus, MSI, Evening classes
8	1.1	4.08	61.35	Special focus, MSI, Award level
9	1.05	3.88	65.23	MSI, Special focus, System
10	1.01	3.74	68.97	Special focus, Award level, Number of online programs
11	0.99	3.66	72.64	Special focus, MSI, Award level
12	0.88	3.24	75.88	Special focus, MSI, Award level

Table 3. Cluster Validation Statistics for Institution-Level and CBSA-Level HCA

k	Silhouette	Min_n	Max_n
Institutions			
2	0.185	846	1534
3	0.209	610	924
4	0.217	100	924
5	0.235	100	746
6	0.247	100	746
7	0.273	94	746
8	0.284	14	746
9	**0.295**	**14**	**746**
10	0.263	14	746
11	0.275	14	746
12	0.248	14	492
CBSAs			
2	0.322	191	528
3	0.407	189	339
4	0.472	71	268
5	0.509	71	191
6	0.531	28	191
7	0.549	28	191
8	**0.569**	**23**	**191**
9	0.592	23	191
10	0.604	23	191
11	0.614	23	191
12	0.626	23	191

Note. k = number of clusters; Min_n = minimum cluster size; Max_n = maximum cluster size. Cluster sizes represent the number of institutions or CBSAs in the smallest and largest clusters, respectively.

Table 4. Classes of Postsecondary Institutions Based on Hierarchical Cluster Analysis

Cluster	Description	Example Institutions
1	Four-year HBCUs and HSIs	Alabama A&M University, Bluefield State University, Boricua College, Langston University
2	Large, graduate degree-granting universities	Boise State University, Stephen F. Austin State University, Stony Brook University, University of Arizona
3	Public, two- and four-year broad-access institutions	Contra Costa College, Gulf Coast State College, SUNY Morrisville, Western Nevada College
4	Private, four-year institutions	Hawaii Pacific University, Hood College, Pacific Oaks College, Providence College
5	Medical/health sciences-focused two-year and private four-year colleges	Bellin College, Bryan College of Health Sciences, University of Providence, University of Rio Grande
6	Four-year (bachelor's degree-focused), professions/arts and sciences-focused institutions	Bluffton University, Mars Hill University, Shepherd University, Susquehanna University
7	Two-year, applied and career studies-focused colleges	Athens Technical College, Benjamin Franklin Cummings Institute of Technology, Ohio Institute of Allied Health, Saint Elizabeth College of Nursing
8	Private, four-year (doctoral degree-granting), predominantly STEM-focused universities	Boston College, Harvey Mudd College, Princeton University, University of Southern California
9	Tribal colleges	Bay Mills Community College, College of the Muscogee Nation, Navajo Technical University, Tohono O'odham Community College

Table 5. Distribution of Institutional Clusters Across CBSA Clusters

Panel A. Summary of Institution Types Across CBSA Clusters

		CBSA Cluster							
Institution Cluster		1	2	3	4	5	6	7	8
1	Four-year HBCUs and HSIs	0 (0.00%)	0 (0.00%)	1 (0.29%)	0 (0.00%)	35 (16.75%)	77 (5.92%)	1 (3.12%)	6 (8.82%)
2	Large, graduate degree-granting universities	1 (0.93%)	9 (3.63%)	24 (7.04%)	73 (100.00%)	70 (33.49%)	161 (12.38%)	0 (0.00%)	11 (16.18%)
3	Public, two- and four-year broad-access institutions	0 (0.00%)	230 (92.74%)	71 (20.82%)	0 (0.00%)	83 (39.71%)	354 (27.21%)	0 (0.00%)	8 (11.76%)
4	Private four-year institutions	0 (0.00%)	5 (2.02%)	42 (12.32%)	0 (0.00%)	3 (1.44%)	339 (26.06%)	29 (90.62%)	2 (2.94%)
5	Medical/health sciences-focused two-year and private four-year colleges	0 (0.00%)	1 (0.40%)	33 (9.68%)	0 (0.00%)	0 (0.00%)	59 (4.53%)	0 (0.00%)	1 (1.47%)
6	Four-year (bachelor's degree-focused), professions/arts and sciences-focused institutions	106 (98.15%)	2 (0.81%)	159 (46.63%)	0 (0.00%)	8 (3.83%)	195 (14.99%)	0 (0.00%)	6 (8.82%)
7	Two-year, applied and career studies-focused colleges	0 (0.00%)	0 (0.00%)	3 (0.88%)	0 (0.00%)	0 (0.00%)	66 (5.07%)	2 (6.25%)	29 (42.65%)
8	Private, four-year (doctoral degree-granting), predominantly STEM-focused universities	1 (0.93%)	1 (0.40%)	4 (1.17%)	0 (0.00%)	3 (1.44%)	48 (3.69%)	0 (0.00%)	4 (5.88%)
9	Tribal colleges	0 (0.00%)	0 (0.00%)	4 (1.17%)	0 (0.00%)	7 (3.35%)	2 (0.15%)	0 (0.00%)	1 (1.47%)
CBSAs (n)		88	191	101	71	89	128	23	28
Institutions (n)		108	248	341	73	209	1301	32	68

Panel B. Names of CBSA Clusters

Cluster	Name	Example CBSAs
1	Singular Baccalaureate-Dominant Institution Areas	Amherst Town-Northampton, MA Athens, TN

		Rock Springs, WY Wilmington, OH
2	Minimal Opportunity, Broad-Access Anchor Institution Areas	Amsterdam, NY Hagerstown-Martinsburg, MD-WV Killeen-Temple, TX Wilmington, NC
3	Extensive-Opportunity Higher Education Ecosystems	Cincinnati, OH-KY-IN Grand Rapids-Wyoming-Kentwood, MI Greenville-Anderson-Greer, SC Lakeland-Winter Haven, FL
4	Singular University Hubs	Bozeman, MT Cleveland, MS Troy, AL Whitewater-Elkhorn, WI
5	Broad Access and MSI-Opportunity Areas	Albuquerque, NM Bluefield, WV-VA Duluth, MN-WI Knoxville, TN
6	Robust Higher Education Markets	Los Angeles-Long Beach-Anaheim, CA New York-Newark-Jersey City, NY-NJ St. Louis, MO-IL Washington-Arlington-Alexandria, DC-VA-MD-WV
7	Private College Hubs	Barre, VT Bridgeport-Stamford-Danbury, CT Gainesville, GA Sioux Falls, SD-MN
8	Two-Year and Career-Focused Opportunity Areas	Beaumont-Port Arthur, TX Bellingham, WA Lake Charles, LA Williamsport, PA

Note. Panel A displays the percent of institutions of each institutional cluster within each CBSA cluster.

Table 6. CBSA Descriptive Characteristics by CBSA-Level Cluster

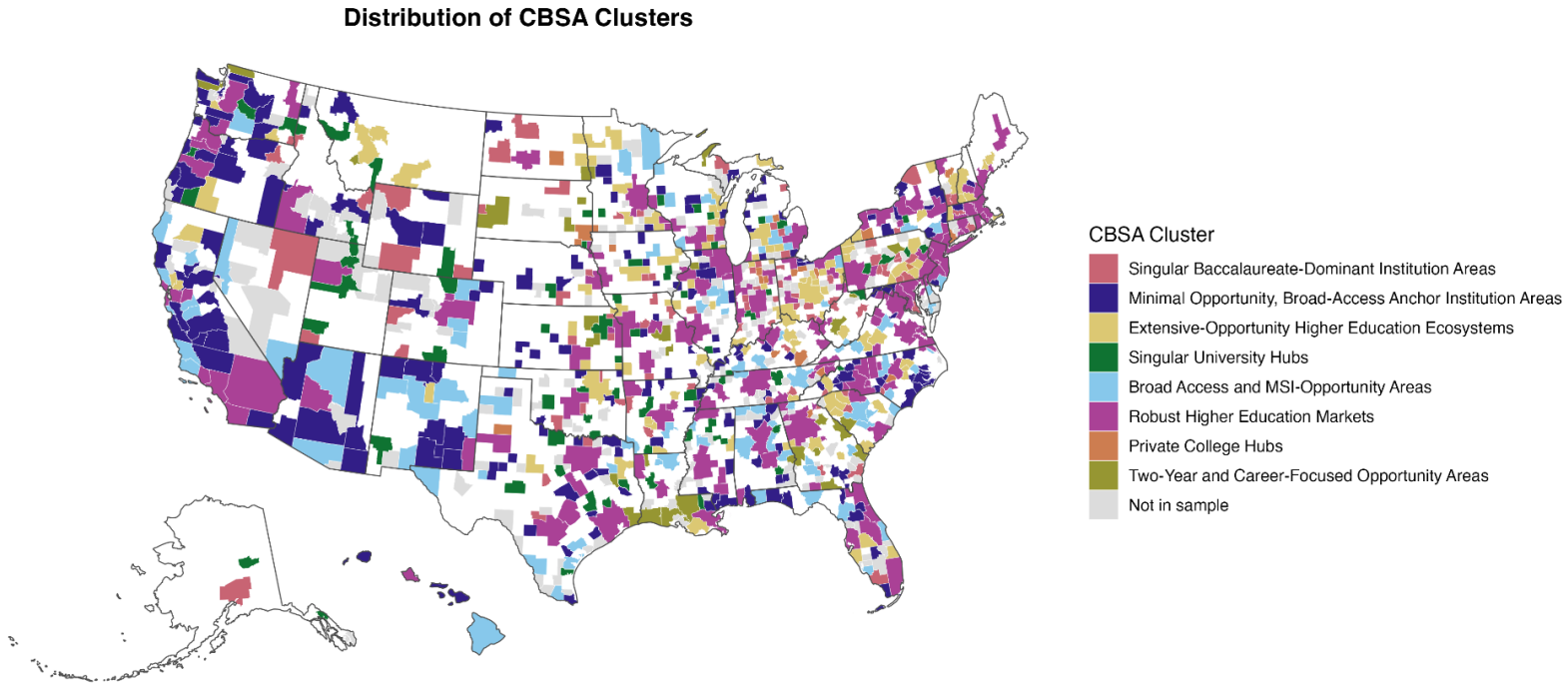
	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8
CBSAs (n)	88	191	101	71	89	128	23	28
CBSA Type								
Micropolitan	0.75	0.71	0.32	0.65	0.27	0.09	0.74	0.43
Metropolitan	0.25	0.29	0.68	0.35	0.73	0.91	0.26	0.57
Number of Institutions	1.23 (0.66)	1.30 (1.44)	3.38 (2.51)	1.03 (0.17)	2.35 (0.97)	10.16 (13.47)	1.39 (0.78)	2.43 (1.48)
Total Population (100,000s)	0.81 (0.82)	1.54 (4.36)	2.68 (3.61)	1.05 (1.20)	3.05 (3.25)	15.64 (26.35)	1.25 (1.93)	1.90 (2.03)
Population Density	104.40 (100.95)	97.14 (105.04)	195.72 (163.82)	92.29 (82.78)	184.59 (173.62)	459.66 (505.64)	193.58 (294.21)	107.52 (90.08)
Bachelor's Degree or Higher (%)	25.70 (7.59)	23.55 (7.88)	28.87 (7.61)	33.93 (9.25)	29.93 (9.10)	34.29 (9.29)	26.97 (8.93)	28.80 (7.52)
Median Household Income ($)	68,703.23 (12,065.51)	67,334.59 (15,094.24)	70,084.37 (11,846.64)	65,262.63 (13,641.16)	67,717.99 (14,420.10)	77,608.11 (16,450.74)	71,330.17 (13,893.78)	66,245.54 (8,433.64)
Below the Poverty Level (%)	8.97 (3.02)	10.61 (3.95)	9.03 (3.00)	9.65 (4.14)	11.46 (4.90)	8.90 (2.79)	8.65 (4.12)	10.07 (3.33)
Unemployment Rate (%)	2.56 (0.80)	2.95 (0.94)	2.76 (0.61)	2.94 (0.87)	3.34 (0.92)	3.05 (0.65)	2.45 (0.71)	2.70 (0.79)
White Population (%)	81.90 (10.90)	67.65 (20.03)	75.72 (13.54)	73.70 (15.15)	57.25 (21.42)	64.65 (16.88)	80.14 (16.67)	72.07 (15.80)
Black Population (%)	4.49 (6.72)	7.13 (10.98)	8.82 (10.72)	7.29 (11.66)	13.93 (16.41)	12.65 (11.22)	3.25 (4.24)	13.49 (14.46)
Hispanic or Latino Population (%)	7.26 (6.03)	17.20 (18.08)	8.12 (7.67)	10.56 (11.95)	19.36 (22.63)	13.30 (12.08)	10.87 (13.68)	7.39 (5.96)
Asian Population (%)	1.08 (1.05)	1.89 (4.28)	1.88 (1.94)	2.15 (1.65)	2.66 (2.82)	4.16 (5.14)	1.71 (2.22)	1.67 (1.15)

Table 7. Multinomial Logistic Regression Predicting CBSA Cluster Membership

	Cluster 2		Cluster 3		Cluster 4		Cluster 5		Cluster 6		Cluster 7		Cluster 8	
	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.	Est.	S.E.
Total Population	10.64***	3.05	10.72***	3.05	5.50	3.58	10.25***	3.06	12.06***	3.04	3.84	4.19	11.23***	3.22
Population Density	-1.67**	0.54	0.17	0.51	-1.08	0.75	-0.25	0.53	0.36	0.51	1.23*	0.62	-1.78	0.92
BA or Higher	-0.66*	0.28	0.38	0.29	1.77***	0.29	1.24***	0.29	0.92**	0.30	0.05	0.45	0.79*	0.39
Median Household Income	-0.08	0.34	-0.95*	0.38	-1.91***	0.42	-1.28***	0.38	-1.35***	0.40	-0.54	0.54	-1.17*	0.56
Poverty Rate	0.06	0.27	-0.43	0.34	-0.28	0.36	-0.03	0.33	-0.22	0.36	0.03	0.47	-0.35	0.48
Unemployment Rate	0.05	0.18	-0.02	0.21	0.18	0.21	0.37	0.20	-0.05	0.22	-0.50	0.33	-0.21	0.30
Black	0.37	0.24	0.51	0.27	0.44	0.29	0.81**	0.27	0.68*	0.27	-1.13	0.83	0.91**	0.32
Hispanic or Latino	0.78**	0.28	0.05	0.36	0.85**	0.33	1.22***	0.30	0.57	0.34	0.66	0.40	0.21	0.53
Asian	1.03	0.58	0.85	0.60	1.06	0.63	1.15	0.59	1.00	0.59	0.79	0.67	1.01	0.70
Intercept	3.22***	0.75	3.02***	0.75	1.14	0.86	2.74***	0.75	3.10***	0.75	-0.41	1.09	1.41	0.81

Note. $N = 719$; * $p < .05$, ** $p < .01$, *** $p < .001$

Figure 1. Geographic Distribution of U.S. Core-Based Statistical Areas and CBSA Clusters



Note. CBSAs shaded in gray were not included in the sample because they either did not contain any of the sample postsecondary institutions.

Appendices

Appendix A. Institutional Characteristics by Institution-Level Cluster

Variable	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8	Cluster 9
Institutions (n)	120	349	746	420	94	476	100	61	14
Institution Type									
Public 2-year	0.06	0.00	0.79	0.01	0.10	0.00	0.34	0.00	0.43
Private 2-year	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.00	0.00
Public 4-year	0.49	0.99	0.21	0.03	0.12	0.26	0.05	0.15	0.43
Private 4-year	0.45	0.01	0.00	0.96	0.79	0.74	0.12	0.85	0.14
System									
Not in system	0.55	0.30	0.39	0.95	0.55	0.82	0.48	0.90	0.93
In system	0.45	0.70	0.61	0.05	0.45	0.18	0.52	0.10	0.07
Award Level									
Associate	0.16	0.00	1.00	0.02	0.47	0.07	0.99	0.00	0.93
Baccalaureate	0.68	0.00	0.00	0.03	0.19	0.90	0.01	0.15	0.07
Master's	0.08	0.45	0.00	0.42	0.10	0.02	0.00	0.07	0.00
Doctoral	0.08	0.55	0.00	0.53	0.24	0.00	0.00	0.79	0.00
Special Focus									
Not special focus	0.23	0.58	0.66	0.23	0.00	0.14	0.05	0.56	0.43
Professions-focused	0.50	0.42	0.34	0.71	0.00	0.46	0.04	0.00	0.50
Applied and career studies	0.00	0.00	0.00	0.00	0.00	0.00	0.74	0.00	0.00
Arts and sciences	0.27	0.00	0.00	0.00	0.00	0.40	0.05	0.00	0.07
Business	0.00	0.00	0.00	0.05	0.00	0.00	0.02	0.00	0.00
Medical/health sciences	0.00	0.00	0.00	0.00	1.00	0.00	0.10	0.00	0.00
STEM	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.44	0.00
Number of degree programs	47.77 (31.14)	199.63 (119.10)	89.35 (47.11)	95.29 (62.97)	47.90 (44.00)	54.58 (35.59)	42.76 (52.58)	207.89 (143.41)	21.64 (20.10)
Number of online programs	0.06 (0.10)	0.13 (0.14)	0.11 (0.13)	0.23 (0.28)	0.16 (0.21)	0.09 (0.16)	0.14 (0.26)	0.06 (0.11)	0.27 (0.36)
Evening									
No	0.33	0.47	0.28	0.31	0.40	0.68	0.43	0.52	0.64
Yes	0.67	0.53	0.72	0.69	0.60	0.32	0.57	0.48	0.36

Instructional expenditures per FTE	9312.38 (4475.71)	12379.99 (6395.46)	7855.89 (3082.21)	10445.20 (4864.54)	11601.46 (7074.23)	13186.11 (8349.93)	9148.76 (5599.91)	41043.69 (36257.63)	15066.86 (12342.01)
Tuition	10569.29 (6800.59)	9005.01 (4028.07)	3308.75 (1659.78)	35170.80 (13942.16)	20966.77 (12395.23)	32260.15 (19789.33)	12644.53 (9250.13)	52933.54 (20634.24)	3433.29 (2282.96)
MSI									
None	0.08	0.78	0.67	0.83	0.90	0.95	0.84	0.97	0.00
HBCU	0.72	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00
HSI	0.19	0.22	0.33	0.17	0.10	0.05	0.13	0.03	0.00
TCU	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Adult students	0.22 (0.19)	0.14 (0.11)	0.28 (0.10)	0.18 (0.20)	0.39 (0.22)	0.10 (0.11)	0.38 (0.24)	0.05 (0.11)	0.49 (0.20)
Pell recipients	0.59 (0.15)	0.33 (0.11)	0.30 (0.11)	0.33 (0.13)	0.41 (0.15)	0.31 (0.12)	0.44 (0.21)	0.19 (0.08)	0.49 (0.19)
Proportion of non-White students	0.89 (0.17)	0.46 (0.21)	0.53 (0.22)	0.48 (0.19)	0.46 (0.20)	0.38 (0.16)	0.52 (0.27)	0.55 (0.19)	0.91 (0.15)
Size	4021.97 (4058.78)	22205.49 (17871.95)	11112.05 (11602.15)	6567.48 (15640.73)	2649.05 (4164.69)	2584.73 (3728.97)	2420.39 (3290.39)	16509.28 (14804.36)	896.71 (671.92)
Proportion of undergraduate students	0.93 (0.09)	0.78 (0.10)	1.00 (0.01)	0.67 (0.16)	0.85 (0.20)	0.93 (0.09)	1.00 (0.01)	0.60 (0.20)	0.99 (0.02)
Selectivity	0.80 (0.23)	0.78 (0.19)	1.00 (0.02)	0.78 (0.16)	0.83 (0.22)	0.73 (0.24)	0.95 (0.14)	0.33 (0.30)	0.99 (0.03)